

Quiz 1

ECON 308 – International Economic Relations

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1. Suppose there are two countries, **Home** and **Foreign**, and two goods, **Wine** and **Cheese**. The unit labor requirements are:

Country	Wine (hours per unit)	Cheese (hours per unit)
Home	2	1
Foreign	6	2

Each country has 120 units of labor.

- Compute the **maximum production possibilities** (Wine and Cheese) for each country.
 - Calculate the **opportunity cost** of producing Wine in terms of Cheese for both countries.
 - Identify which country has a **comparative advantage** in Wine and which in Cheese.
 - If the world relative price of Wine is $P_W/P_C = 1.5$, determine which good each country will export.
2. Suppose trade flows follow the basic gravity equation:

$$T_{ij} = A \cdot \frac{Y_i Y_j}{D_{ij}}$$

where T_{ij} is trade between country i and j , Y denotes GDP, D is the distance between countries, and $A = 1$.

The data are as follows:

- Country A GDP: \$2,000 billion
- Country B GDP: \$1,500 billion
- Country C GDP: \$500 billion

- Distance between A and B: 1,000 km
 - Distance between A and C: 5,000 km
- (a) Calculate the predicted trade flow T_{AB} and T_{AC} .
- (b) Find the ratio of trade between A–B versus A–C.
- (c) Suppose a free trade agreement between A and C effectively **halves the distance cost** (so effective distance = 2,500 km). Recalculate T_{AC} and explain how the agreement changes trade flows.